

ZUERICH-FLUNTERN, Switzerland

WMO: 066600

Lat: 47.3781N

Lon: 8.5658E

Elev: 1829

StdP: 13.75

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.4	20.8	12.2	10.9	19.9	15.4	12.9	23.3	26.8	46.2	23.6	46.5	4.8	50	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	85.6	67.2	82.0	65.8	78.9	64.5	68.6	81.4	67.2	78.8	65.8	76.2	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	97.1	73.0	63.1	92.6	71.3	62.0	88.8	70.2	33.8	81.7	32.6	79.1	31.5	76.6	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.1	15.5	12.3		13.4	90.4	4.9	3.0	9.8	92.6	6.9	94.3	4.2	96.0	0.6	98.2	
				12.6	71.6	4.8	1.7	9.1	72.9	6.3	73.9	3.6	74.8	0.1	76.1	
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	33.8	35.8	43.0	49.9	57.2	63.8	66.8	66.1	58.6	51.3	41.4	35.3	(6)
(7)		DBStd	13.67	7.38	7.91	7.48	7.49	6.76	7.11	6.10	6.12	6.02	6.70	6.80	7.00	(7)
(8)		HDD50	2046	503	399	236	95	15	1	0	0	6	67	267	457	(8)
(9)		HDD65	5660	967	819	683	455	256	107	52	60	205	426	708	922	(9)
(10)		CDD50	2165	1	1	19	91	237	416	521	500	264	106	9	0	(10)
(11)		CDD65	302	0	0	0	0	13	72	108	95	14	0	0	0	(11)
(12)		CDH74	2072	0	0	0	9	104	527	748	623	61	0	0	0	(12)
(13)		CDH80	528	0	0	0	0	11	138	204	173	2	0	0	0	(13)
(14)	Wind	WSAvg	4.5	5.2	5.2	5.6	4.7	4.5	4.1	4.1	3.6	3.9	3.8	4.4	5.1	(14)
(15)	Precipitation	PrecAvg	44.1	2.6	2.6	2.8	3.4	4.6	5.0	4.9	5.1	3.6	3.0	3.2	3.2	(15)
(16)		PrecMax	61.0	6.1	11.4	9.3	7.9	11.3	9.1	11.2	11.6	9.0	7.4	8.3	6.7	(16)
(17)		PrecMin	33.9	0.0	0.4	0.5	0.2	0.7	1.3	1.5	1.1	0.9	0.3	0.0	0.0	(17)
(18)		PrecStd	6.2	1.5	1.8	1.7	1.8	1.9	2.0	2.1	2.2	1.9	1.7	1.8	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	56.7	66.0	75.0	81.1	88.8	90.0	90.1	79.2	70.3	60.8	53.0	(19)
(20)		MCWB	46.8	45.9	50.8	56.0	62.6	67.9	68.0	67.8	64.4	59.1	51.9	46.4		(20)
(21)		2%	DB	50.1	52.7	61.7	70.6	77.0	84.0	85.4	84.5	75.7	66.2	56.5	50.0	(21)
(22)		MCWB	44.4	44.2	49.0	54.1	60.9	66.1	67.1	67.3	63.3	57.7	50.1	45.4		(22)
(23)		5%	DB	46.6	49.5	57.8	66.5	73.0	80.1	81.6	80.7	71.5	63.0	53.1	47.5	(23)
(24)		MCWB	42.1	42.8	47.7	52.6	58.4	64.9	65.8	66.1	61.2	56.3	48.5	43.8		(24)
(25)		10%	DB	43.7	46.3	54.1	62.6	69.3	76.2	78.1	76.8	68.0	60.0	50.2	45.0	(25)
(26)		MCWB	40.3	41.0	45.9	50.7	57.0	63.2	64.4	64.7	60.1	55.0	46.8	41.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	47.6	52.6	57.7	64.5	70.7	70.6	70.3	66.4	60.9	53.1	48.3	(27)
(28)		MCDWB	52.7	54.3	62.4	72.0	78.0	85.1	85.1	83.1	76.4	66.9	58.3	50.8		(28)
(29)		2%	WB	45.0	45.0	50.4	55.1	62.0	67.7	68.3	68.5	64.3	59.0	50.9	46.0	(29)
(30)		MCDWB	48.8	51.3	58.6	67.3	73.7	79.8	81.1	81.2	72.8	64.3	55.1	49.1		(30)
(31)		5%	WB	42.5	43.2	48.4	53.4	60.0	66.0	66.9	67.0	62.5	57.3	49.0	44.1	(31)
(32)		MCDWB	46.0	48.6	56.2	64.5	70.0	77.2	78.5	78.3	69.5	61.9	52.6	47.2		(32)
(33)		10%	WB	40.4	41.3	46.5	51.5	58.1	64.3	65.5	65.6	60.8	55.6	47.1	42.0	(33)
(34)		MCDWB	43.5	45.9	53.2	61.0	67.0	74.4	75.8	74.9	66.8	59.0	50.0	44.7		(34)
(35)	Mean Daily Temperature Range		MDBR	7.5	10.2	13.5	15.8	15.9	16.3	16.2	15.5	13.6	11.1	8.2	7.2	(35)
(36)		5% DB	MCDBR	11.2	15.8	19.6	22.6	22.2	22.3	21.9	21.7	18.8	15.2	12.8	10.1	(36)
(37)			MCWBR	7.5	9.2	10.1	10.4	9.9	8.9	8.3	8.2	8.4	8.0	7.8	6.8	(37)
(38)			MCDBR	10.1	13.9	17.4	20.7	19.7	20.3	20.0	19.7	16.8	13.9	11.8	9.3	(38)
(39)		5% WB	MCWBR	7.4	8.5	9.4	9.9	9.4	8.6	8.2	8.1	8.0	7.8	7.6	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.287	0.311	0.346	0.380	0.394	0.399	0.401	0.393	0.368	0.342	0.311	0.283	(40)	
(41)		taud	2.445	2.381	2.347	2.291	2.296	2.322	2.309	2.346	2.400	2.459	2.492	2.456	(41)	
(42)		Ebn at Noon	256	271	277	277	276	275	272	269	266	256	244	243	(42)	
(43)		Edh at Noon	21	27	33	39	40	39	39	36	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	378	648	1032	1377	1577	1813	1757	1533	1153	728	409	311	(44)	
(45)		RadStd	42	87	139	200	159	168	160	156	118	79	50	47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (119 neighbors)	N/A	N/A	N/A	+1.28	+0.76	+0.61	N/A	N/A	+103	+29	(47)

Nomenclature: See separate page